2026

$2 + 0 + 2 + 6 = 10 \to 1$

so 2026 leaves remainder 1 when divided by 9

Numbers That Talk

Divisibility shortcuts, and the modular arithmetic hiding behind them

Zhiheng Li · Yueran

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Is 2026 divisible by 9?

Magic trick 1: the vanishing digit

1. Think of a 4-digit number (not all digits the same).
2. Shuffle its digits into a new number.
3. Subtract: bigger minus smaller.
4. Circle one non-zero digit of the result. Hide it.
5. Read me the other digits, in any order.

example

5274 → **7425**

7425 − 5274 = **2 1 5 1**

They read: 2, 1, 1 → sum 4

4 needs 5 to reach 9. The hidden digit is 5.

If the sum is already a multiple of 9, the answer is 9.

A clock with only 9 marks



walk 10 steps = walk 1 step

$$2026 = 2\cdot 10^3 + 0\cdot 10^2 + 2\cdot 10 + 6$$

on the 9-clock every 10^k is just 1

$$2026 \equiv 2 + 0 + 2 + 6 = 10 \equiv 1 \pmod 9$$

Same digits $\Longrightarrow$ same spot on the clock.

Subtract $\Longrightarrow$ land on the top, that is 0 (mod 9).

So the digits of the result add to a multiple of 9. You read all but one; I fill in the gap.

Bigger chunks: 99 and 999

$100 = 99 + 1 \implies 100 \equiv 1 \pmod{99}$

add 2-digit chunks → remainder mod 99 (and 11, 33)

$1000 = 999 + 1, \quad 999 = 27 \times 37$

add 3-digit chunks → tests 27 and 37

12 **34** **53**

12 + 34 + 53 = 99 ✓
so 99 | 123453

45 **658**

45 + 658 = 703 = 19 × 37 ✓

45658

37 divides 45658, no long division needed. (It is 37 × 1234. We checked.)
Same trick, 2026 → 2 + 026 = 28, so 2026 ≡ 28 (mod 37).

Always chunk from the right: 45 | 658, not 456 | 58

Family one: $10^k \equiv 1$

$10 \equiv 1 \pmod 9$ → digit sum

$100 \equiv 1 \pmod{99}$ → 2-digit chunks

$1000 \equiv 1 \pmod{999 = 27 \times 37}$ → 3-digit chunks

7 ? 13 ?

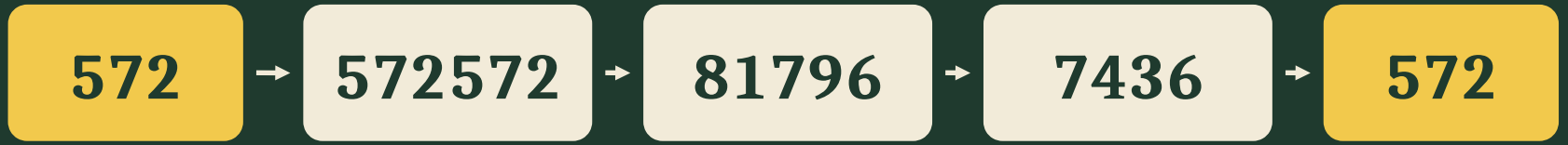
999 = 27 × 37. No 7, no 13 anywhere.

Does 7 even have a rule?

What if $10^k \equiv -1$ instead?

over to Yueran

Magic trick 2: 1001 brings it back



Every division comes out exact. The last one gives back the number you started with. Try your own number.

live

_____ → _____ → ÷ 7 → ÷ 11 → ÷ 13 → _____

Why it works: 1000 acts like −1

$$abcabc = abc \cdot 1000 + abc = abc \cdot 1001 = abc \cdot 7 \cdot 11 \cdot 13$$

one digit at a time: 10 = 11 − 1, so $10 \equiv -1 \pmod{11}$

$10^3 \equiv -1$	$10^2 \equiv +1$	$10 \equiv -1$	1
2	0	2	6
−	+	−	+

$6 - 2 + 0 - 2 = 2 \implies 2026 \equiv 2 \pmod{11}$ ✓

start the + from the units digit

3 digits at a time: 1000 = 1001 − 1, so $1000 \equiv -1$

2 − **026** = 24

24 ≡ 3 (mod 7), 24 ≡ 11 (mod 13). Both are the true remainders of 2026.

572572 → 572 − 572 = 0 ⟹ 7, 11, 13 all divide it.

One idea, every rule

find k with $10^k \equiv +1$ or $-1 \pmod d$

+1 → cut into k-digit chunks from the right and add them

−1 → same chunks, alternate + − + −

The result is the remainder mod d, not just yes/no.

d	k	$10^k \equiv$	rule
3, 9	1	+1	digit sum
11	1	−1	alternate digits
101	2	−1	alternate 2-chunks
37 (and 27)	3	+1	add 3-chunks
7, 13 (and 77)	3	−1	alternate 3-chunks
41	5	+1	add 5-chunks

2026: 2+0+2+6 = 10 → mod 9 = 1 · 6−2+0−2 → mod 11 = 2 · 026−2 = 24 → mod 7 = 3, mod 13 = 11 · 2+026 → mod 37 = 28

Why every prime gets a rule

Fermat: $10^{p-1} \equiv 1 \pmod{p}$ for every prime $p \neq 2, 5$

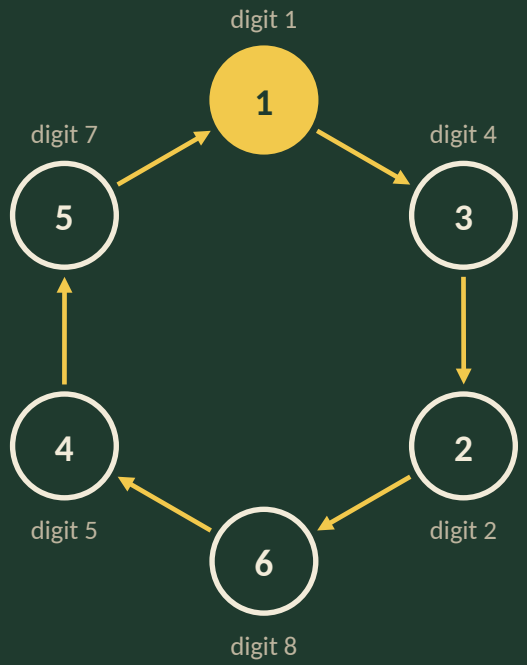
So (p − 1)-digit chunks always work.

Shortest additive chunk = $\text{ord}_p(10)$, the order of 10 mod p.

And that same number is the period of 1/p.

$1/7 = 0.142857\,142857\ldots \quad \text{ord}_7(10) = 6$

142857 × 2 = 285714, × 3 = 428571, × 6 = 857142. Same digits, rotated. 10 is a primitive root mod 7.



remainders of 1 ÷ 7 = powers of 10 mod 7

If you know what 10 is mod d, you can build the rule for d yourself.

Fermat says the rule exists. The order of 10 says how short it can be.

In the full deck: chop-the-last-digit rule for 7 (x − 2y), why ISBN uses 11, Midy's 142 + 857 = 999, and every proof.
github.com/ZhihengLi0/math5248

Thank you. Now go try the 1001 trick on a friend.